Research Statement

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I am currently pursuing my Master's in Data Science and Artificial Intelligence at Saarland University and working as a student research assistant in Mario Fritz's Trustworthy Machine Learning group at CISPA. My research interests are broadly in AI safety, with a focus on evaluating LLMs and long-term personalization and memory.

My work on LLM-assisted peer review [1] studied how author-related information can introduce bias into the review process under single-blind conditions. We investigated whether information about authors influences LLM-generated reviews and their assessments of scientific papers. This work was published in Findings of ACL 2026, and I presented it at ACL 2026 in San Diego. In my more recent work [2], I investigate linguistic bias in LLM-assisted peer review under double-blind conditions. We developed a controlled framework to vary linguistic confidence and language proficiency while preserving the scientific content of research papers. Our experiments show that LLM reviewers favor papers written with higher confidence, while the effects of language proficiency are smaller and vary across models. We also found that changes in linguistic style can affect review scores and acceptance recommendations.

My work has also focused on evaluating LLM safety for real users [3]. We studied how LLMs provide financial and health advice to users with different levels of vulnerability. We found that safety evaluations can miss important risks when they do not consider the user's circumstances. We also found that including realistic user information in prompts does not necessarily make the responses safer. This work was accepted at IASEAI 2026. I also participated in an interdisciplinary research incubator at AI Safety Saarland, where we studied gender bias in LLM-generated stories, including the representation of occupations and social roles.

My Master's thesis, *Long-Horizon Personalization via Personal Knowledge Graphs*, focuses on how language models can maintain and use personal information across extended conversations. I investigate how personal knowledge graphs can represent user-specific information, update it as new information becomes available, and retrieve relevant information for future interactions. This includes handling information that changes over time and deciding what should be retained, updated, or treated as outdated.

I am also a research fellow in the Fall 2026 cohort of SPAR (Supervised Program for Alignment Research), where my current work investigates moral shadowbanning in conversational AI. We study whether models provide different levels of assistance depending on their perceptions of a user's moral character.

For my PhD, I am interested in the safety, privacy, and reliability of AI agents, particularly agents that interact with users over long periods and maintain information about them. Two questions I would like to explore are how personalized agents should use information across different contexts and how we can evaluate their safety in realistic interactions.

One question I would like to investigate is **when a personalized agent should use or share information it remembers about a user**. My Master's thesis focuses on representing and retrieving personal information, but information being available or relevant does not necessarily mean it is appropriate to use. An agent may remember something from a private conversation that should not be disclosed when communicating with another agent or completing a different task. I

am interested in studying how agents can decide when it is appropriate to use or share personal information, depending on the situation and who receives it. The COMPASS group's work on contextual safety in agent-to-agent conversations [4] is particularly relevant to this question. I would like to explore how contextual integrity can be evaluated across long interactions where an agent accumulates personal knowledge over time.

I am also interested in **evaluating agent safety in realistic, context-dependent settings**. My previous research on LLM-assisted peer review and safety evaluation has shown how important context can be when evaluating LLMs. For AI agents, I would like to study evaluations that consider not only whether a task was completed, but also how it was completed. For example, an agent might successfully carry out a task while sharing private information with another agent or taking an action the user did not authorize. I am interested in developing evaluation settings that account for interaction history, user intent, and information-sharing boundaries, particularly in multi-turn and multi-agent environments.

The COMPASS group's research on contextual integrity, AI agent security, and safety evaluation closely matches these interests. I would like to build on my experience in LLM evaluation and long-term personalization while studying the privacy and security questions that arise when agents retain information and act across different contexts. These are the directions I hope to develop further during my PhD.

References

[1] S. S. M. Vasu, I. Sheth, H.-P. Wang, R. Binkyte, and M. Fritz. "Justice in Judgment: Unveiling (Hidden) Bias in LLM-assisted Peer Reviews." *Findings of the Association for Computational Linguistics: ACL 2026*.

[2] "Style over Substance? Linguistic Bias in LLM-Based Peer Review." Manuscript under review at ICLR 2027.

[3] M. Kempermann, S. S. M. Vasu, M. Raveenthiran, T. Farrell, and I. Weber. "Safe for Whom? Rethinking How We Evaluate the Safety of LLMs for Real Users." *Proceedings of the IASEAI Conference*, 2026.

[4] "ConVerse: Benchmarking Contextual Safety in Agent-to-Agent Conversations." *Findings of EACL*, 2026.